Homework

(1) Suppose that the change in the value of a portfolio over a 10-day time horizon is normally distributed with a mean of zero and a standard deviation of 20 million. Compute the 10-day 99% VaR and ES for the portfolio.

(2) Consider two independent random variables L_1 and L_2 with the following distributions:
- $P(L_1 = 0) = 0.98$, $P(L_1 = 10) = 0.02$
- $P(L_2 = 0) = 0.98$, $P(L_2 = 10) = 0.02$

(a) Prove that $\mathrm{VaR}_{0.98}$ is not subadditive.
(b) Determine whether $\mathrm{VaR}_{0.99}$ is subadditive.
(c) Find $\mathrm{ES}_{0.98}$ for $L_1, L_2, L_1 + L_2$, respectively.

(3) A bank has two 10 million one-year loans. The probabilities of default are as follows:
- $\mathbb{P}$(Neither loan defaults) = 0.975
- $\mathbb{P}$(Loan 1 defaults, Loan 2 does not default) = 1.25%
- $\mathbb{P}$(Loan 2 defaults, Loan 1 does not default) = 1.25%
- $\mathbb{P}$(Both loans default) = 0%

If a default occurs, all losses between 0% and 100% of the principal are equally likely. If the loan does not default, a profit of 0.2 million is made.
(a) Compute one-year 99% VaR for the Loan 1.
(b) Compute one-year 99% VaR for the bank loan portfolio. Is it subadditive?
(c) Compute the one-year 99% expected shortfall (ES) for the Loan 1 and bank loan portfolio, respectively.

(4) VaR monotonicity says that If $L_1 \leq L_2$, then $\mathrm{VaR}_\alpha(L_1) \leq \mathrm{VaR}_\alpha(L_2)$. Justify the following statement:
If $L_1 < L_2$ almost surely, then $\mathrm{VaR}_\alpha(L_1) < \mathrm{VaR}_\alpha(L_2)$ for any $\alpha < 1$.

(5) Explicitly compute $\mathrm{VaR}_\alpha(L)$ and $\mathrm{ES}_\alpha(L)$, assuming L has the following distributions.
(a) $L \sim \mathrm{Exp}(1/\theta)$ is exponentially distributed with mean θ.
(b) $L \sim \mathrm{Pareto}(1, 1/\xi)$ is Pareto distributed with unit scale and $1/\xi$ shape parameter for $\xi > 0$. Here, L has CDF

$$F(\tau) = \begin{cases} 0 & \tau \leq 1 \\ 1 - \tau^{-\frac{1}{\xi}} & \tau > 1 \end{cases}.$$

How does your answer depend on ξ?